

1. Finding the Right Pitch

A quartet of singing eels is recording a 4-sample take. We model each eel's voice as a single complex exponential, a "pure frequency." Eel k sings at angular frequency $\omega_k = k\omega_0$, where $\omega_0 = \frac{2\pi}{4} = \frac{\pi}{2}$ and $k = 0, 1, 2, 3$. Its unscaled voice has samples

$$\psi_k[n] = e^{ik\omega_0 n}, \quad n = 0, 1, 2, 3.$$

We can also collect these samples into a vector in $\mathbb{C}^4$. For example, eel 2's unscaled voice is

$$\boldsymbol{\psi}_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}.$$

These samples repeat every four steps because $e^{ik\omega_0(n+4)} = e^{ik\omega_0 n}$. The four sample vectors $\{\boldsymbol{\psi}_0, \boldsymbol{\psi}_1, \boldsymbol{\psi}_2, \boldsymbol{\psi}_3\}$ form an orthogonal Fourier basis for period-4 signals, with $\|\boldsymbol{\psi}_k\|^2 = 4$ for every k .

When the eels sing together, their signals add sample by sample. If their respective weights are $\alpha, \beta, \gamma, \delta$, the recorded track is

$$m[n] = \alpha\psi_0[n] + \beta\psi_1[n] + \gamma\psi_2[n] + \delta\psi_3[n].$$

The quartet produces the combined track

$$\mathbf{m} = \begin{bmatrix} m[0] \\ m[1] \\ m[2] \\ m[3] \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -3 \\ 7 \end{bmatrix}.$$

Each entry is the recorded signal's amplitude at a sample time $n = 0, 1, 2, 3$; for example, $m[1] = 3$ is the combined amplitude at sample time 1. Determine how loudly the eel at pitch $\omega_2 = \pi$ (i.e. $k = 2$) is singing in this recording. That is, find the scalar γ such that this eel's contribution to $\mathbf{m}$ is $\gamma\boldsymbol{\psi}_2$.

Answer: Since $\{\boldsymbol{\psi}_0, \boldsymbol{\psi}_1, \boldsymbol{\psi}_2, \boldsymbol{\psi}_3\}$ is an orthogonal basis, $\mathbf{m} = \alpha\boldsymbol{\psi}_0 + \beta\boldsymbol{\psi}_1 + \gamma\boldsymbol{\psi}_2 + \delta\boldsymbol{\psi}_3$ for some scalars, and taking the inner product of both sides with $\boldsymbol{\psi}_2$ isolates γ alone, since every other term vanishes by orthogonality:

$$\langle \mathbf{m}, \boldsymbol{\psi}_2 \rangle = \gamma \langle \boldsymbol{\psi}_2, \boldsymbol{\psi}_2 \rangle = \gamma \|\boldsymbol{\psi}_2\|^2 \implies \gamma = \frac{\langle \mathbf{m}, \boldsymbol{\psi}_2 \rangle}{\|\boldsymbol{\psi}_2\|^2}.$$

With $\mathbf{m} = (1, 3, -3, 7)^\top$ and $\boldsymbol{\psi}_2 = (1, -1, 1, -1)^\top$:

$$\langle \mathbf{m}, \boldsymbol{\psi}_2 \rangle = 1(1) + 3(-1) + (-3)(1) + 7(-1) = 1 - 3 - 3 - 7 = -12, \quad \|\boldsymbol{\psi}_2\|^2 = 4.$$

So $\gamma = -12/4 = -3$: the eel at pitch $\omega_2 = \pi$ is contributing $-3\boldsymbol{\psi}_2$ to the recording, i.e. singing with volume -3 .

2. DTFS Puzzle: What Do We Know So Far? Let x be a real-valued signal satisfying $x[n+4] = x[n]$. We will reveal information about x one clue at a time. For the four-sample representation, use $\omega_0 = \pi/2$ and

$$\boldsymbol{\psi}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{\psi}_1 = \begin{bmatrix} 1 \\ i \\ -1 \\ -i \end{bmatrix}, \quad \boldsymbol{\psi}_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \quad \boldsymbol{\psi}_3 = \begin{bmatrix} 1 \\ -i \\ -1 \\ i \end{bmatrix}.$$

These vectors are orthogonal and have squared norm 4. The DTFS coefficient X_k is the weight of $\boldsymbol{\psi}_k$ in the expansion of $\mathbf{x}$. Taking an inner product with $\boldsymbol{\psi}_k$ isolates that weight:

$$X_k = \frac{1}{4} \langle \mathbf{x}, \boldsymbol{\psi}_k \rangle = \frac{1}{4} \sum_{n=0}^3 x[n] \boldsymbol{\psi}_k^*[n], \quad x[n] = \sum_{k=0}^3 X_k \boldsymbol{\psi}_k[n].$$

In particular, the projection of $\mathbf{x}$ onto the direction $\boldsymbol{\psi}_k$ is the vector

$$\text{proj}_{\boldsymbol{\psi}_k} \mathbf{x} = \frac{\langle \mathbf{x}, \boldsymbol{\psi}_k \rangle}{\|\boldsymbol{\psi}_k\|^2} \boldsymbol{\psi}_k = X_k \boldsymbol{\psi}_k.$$

In each part, use only the clues revealed so far.

- (a) **Clue 1:** $x[0] + x[1] + x[2] + x[3] = 2$. Find X_0 and explain what it measures. Give two different four-sample vectors that satisfy this clue. Is the signal uniquely determined?

Answer: Taking the inner product of $\mathbf{x} = \sum_{k=0}^3 X_k \boldsymbol{\psi}_k$ with $\boldsymbol{\psi}_0$ makes all terms except $X_0 \boldsymbol{\psi}_0$ vanish by orthogonality. Thus

$$\langle \mathbf{x}, \boldsymbol{\psi}_0 \rangle = X_0 \|\boldsymbol{\psi}_0\|^2 = 4X_0.$$

Since $\boldsymbol{\psi}_0 = [1, 1, 1, 1]^\top$, clue 1 gives

$$\langle \mathbf{x}, \boldsymbol{\psi}_0 \rangle = x[0] + x[1] + x[2] + x[3] = 2, \quad X_0 = \frac{2}{4} = \frac{1}{2}.$$

The projected vector is $X_0 \boldsymbol{\psi}_0 = \frac{1}{2} [1, 1, 1, 1]^\top$, the constant component of $\mathbf{x}$. Its weight X_0 is the average sample value. For example, $[2, 0, 0, 0]$ and $[1, 1, 0, 0]$ both satisfy the clue. The signal is not uniquely determined.

- (b) **Clue 2:** $x[0] - x[1] + x[2] - x[3] = 4$. Find X_2 . Use the two clues to determine $x[0] + x[2]$ and $x[1] + x[3]$. Give two different sample vectors satisfying *both* clues. What remains unknown?

Answer: Orthogonality similarly gives $\langle \mathbf{x}, \boldsymbol{\psi}_2 \rangle = 4X_2$. Since $\boldsymbol{\psi}_2 = [1, -1, 1, -1]^\top$ is real, clue 2 gives

$$\langle \mathbf{x}, \boldsymbol{\psi}_2 \rangle = x[0] - x[1] + x[2] - x[3] = 4, \quad X_2 = \frac{4}{4} = 1.$$

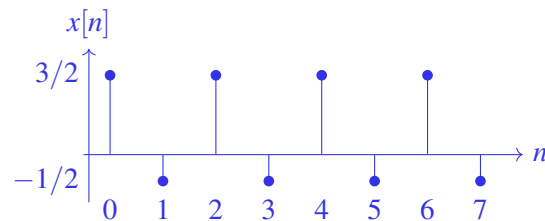
The projected vector is $X_2 \boldsymbol{\psi}_2 = [1, -1, 1, -1]^\top$, the alternating component of $\mathbf{x}$. Adding and subtracting the two clue equations gives $x[0] + x[2] = 3$ and $x[1] + x[3] = -1$. Examples are $[3, -1, 0, 0]$ and $[2, 0, 1, -1]$. We know the sums of the even- and odd-indexed samples, but not how each sum is divided between its two samples. Equivalently, X_1 and X_3 are still unknown.

- (c) **Clue 3:** The projections onto ψ_1 and ψ_3 are both zero. Complete the list X_0, X_1, X_2, X_3 , and reconstruct $x[0], x[1], x[2], x[3]$. Write a formula for $x[n]$ and sketch the samples for $n = 0, 1, \dots, 7$.

Answer: The coefficients are $(1/2, 0, 1, 0)$. Thus

$$x[n] = \frac{1}{2} + (-1)^n, \quad [x[0], x[1], x[2], x[3]] = [3/2, -1/2, 3/2, -1/2].$$

A two-period sketch alternates heights $3/2$ and $-1/2$ at integer samples, starting at $3/2$ when $n = 0$:



The plot shows two four-sample blocks (four fundamental periods).

- (d) Why did the first two clues allow multiple answers, while all three clues determine a unique signal? What is the fundamental period of the reconstructed signal? Does that contradict the statement $x[n+4] = x[n]$?

Answer: The first two clues fix only two Fourier coefficients. The third fixes the remaining two; coefficients in a basis uniquely specify a signal. The fundamental period is 2. A two-periodic signal is also four-periodic, so there is no contradiction.